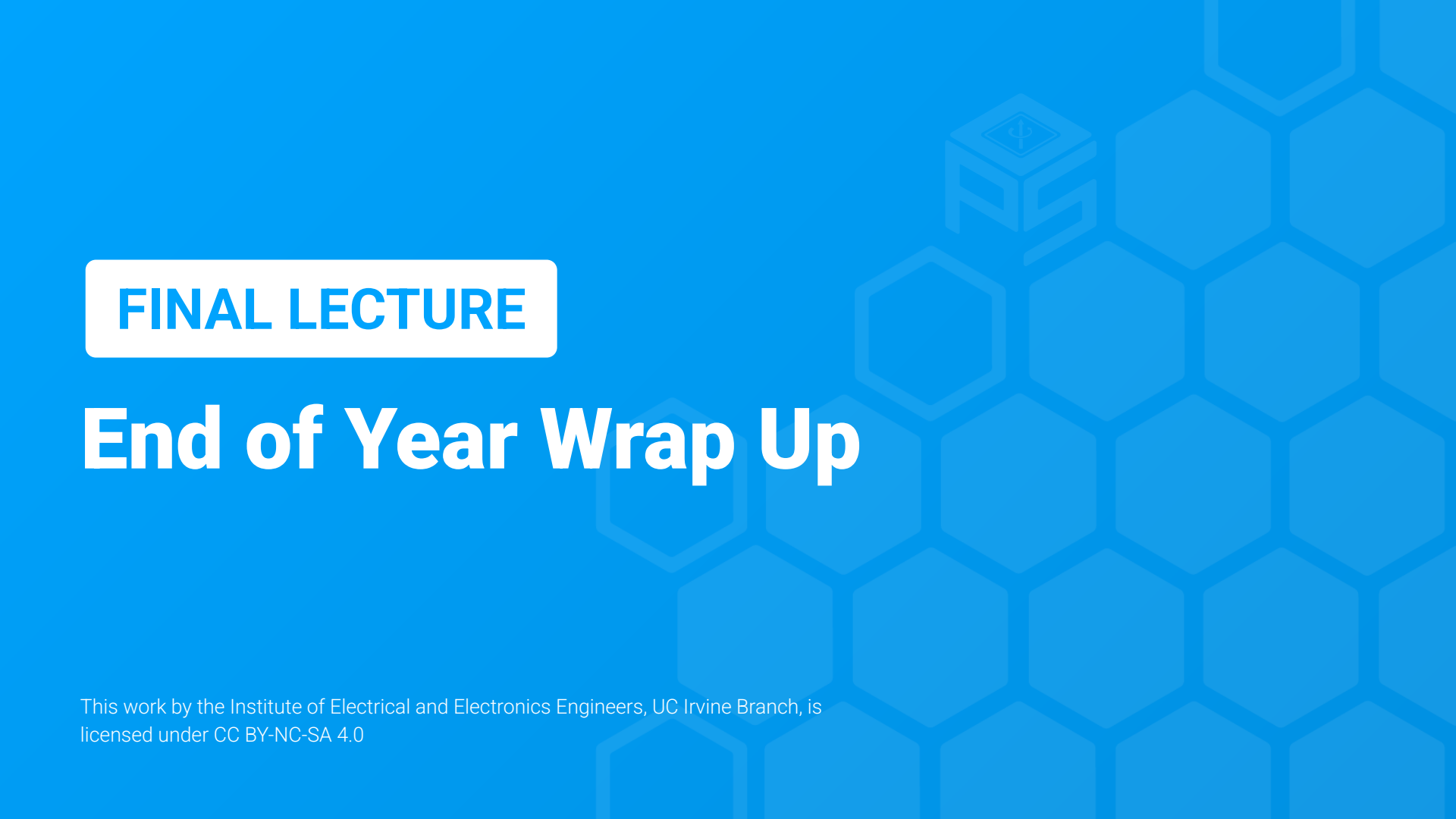




FINAL LECTURE

End of Year Wrap Up

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SECTION I

Circuits Review

Circuits Reflection

- What are the core concepts of circuits that you should take away from OPS?
- What pitfalls are important to look out for when trying to implement a circuit?



Voltage

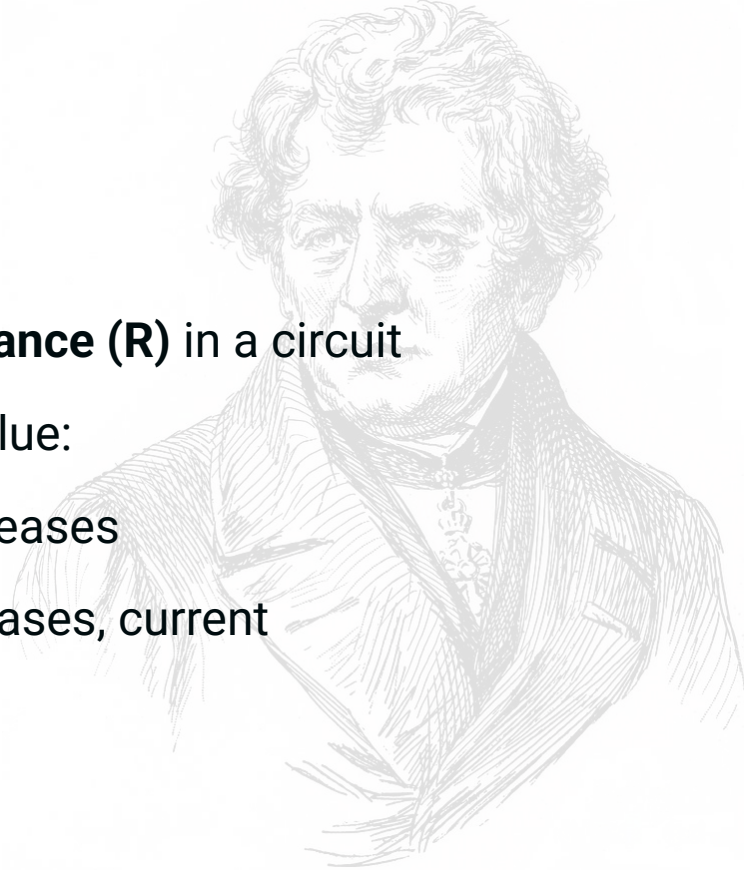
- **Voltage** is the **electric potential difference**
 - Current can flow where **voltage** exists
 - Measured in **Volts, V**
 - Ex. **Batteries** are a voltage source
- **Requires a reference point** because it measures the difference in volts between two points



Ohm's Law

$$V = IR$$

- Relates the **voltage (V)**, **current (I)**, and **resistance (R)** in a circuit
- Suppose we hold the voltage at a constant value:
 - As the resistance increases, current decreases
 - In the opposite case, as resistance decreases, current increases



SECTION II

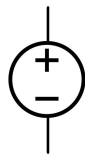
Circuit Analysis Review

Schematic Symbols

Battery



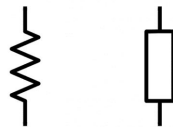
Voltage Source



Ground



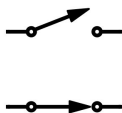
Resistor



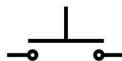
Potentiometer



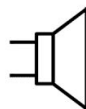
Switches



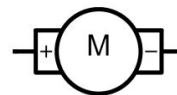
Pushbutton



Speaker



DC Motor



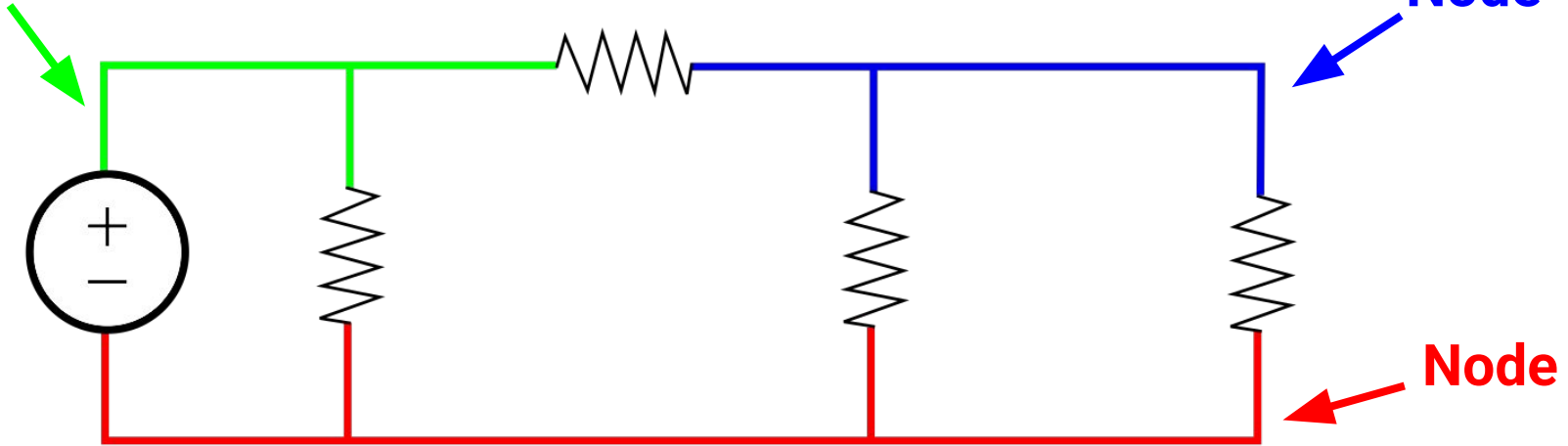
LED



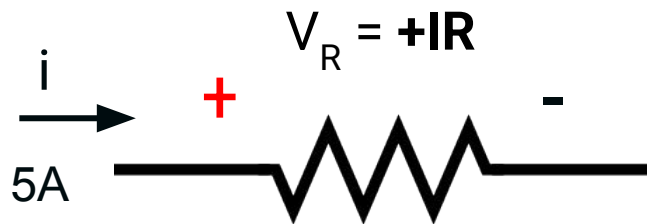
Nodes

- A **node** is a **connection** between two or more **components**
- All points on the **same node** have the **same voltage**

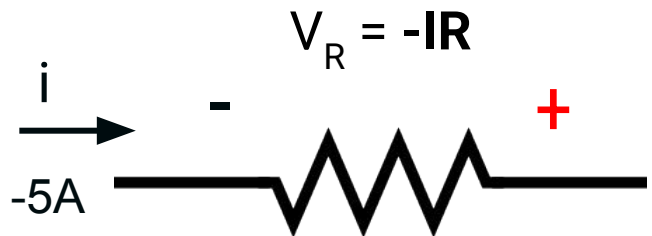
Node



Voltage Drop Sign Convention (Cont'd)



- The sign of the voltage drop is **positive** if current flows from $+$ to $-$ terminals
 - Conventional current flows from $+$ to $-$ terminals

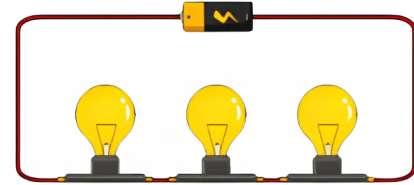


- The sign of the voltage drop is **negative** if current flows $-$ to $+$ terminals

Series and Parallel Circuits

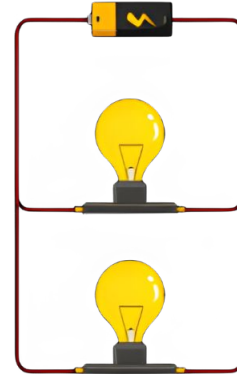
Series

- Two or more elements are in **series** if they exclusively are connected in one line
- Elements in series carry the **same current**



Parallel

- Two or more elements are in **parallel** if they are connected to they share the same two nodes
- Elements in parallel have the **same voltage** across them

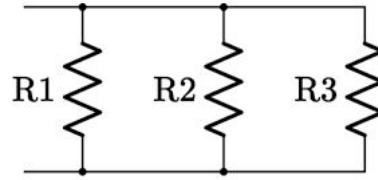


Series and Parallel Circuits

- If **multiple resistors** are arranged in **series or parallel**, we can treat them as having a **single equivalent resistance**

Parallel

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$



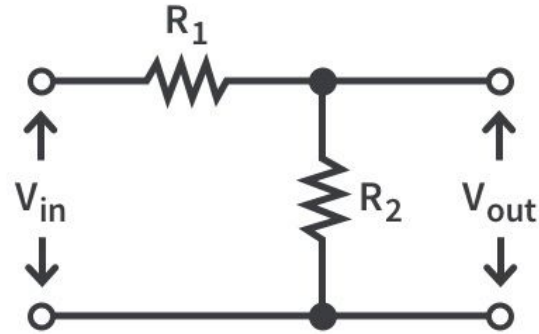
Series

$$R_{eq} = R_1 + R_2 + R_3$$



Voltage Divider

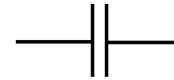
- A circuit configuration with **two resistors** that is used to scale down a voltage (**V_{out}**).
- Creates a ratio of two resistors to achieve a desired output voltage
 - **R₁** and **R₂** are connected in series
 - **V_{in}** is the input voltage
 - **V_{out}** is the output voltage across **R₂**



$$V_{\text{out}} = \left(\frac{R_2}{R_1 + R_2} \right) V_{\text{in}}$$

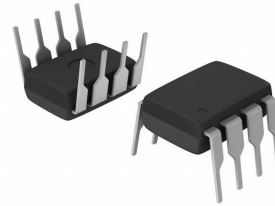
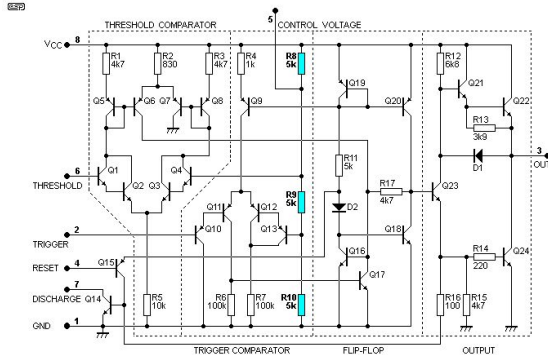
Capacitors

- A **capacitor** stores electrical energy which it charges and discharges
 - The ability of a capacitor to store energy is its **capacitance**, measured in **Farads (F)**
- Unlike a battery, a capacitor can only briefly store a small amount of energy
- When a capacitor is connected to a voltage source, it charges until it reaches the **same voltage** as the voltage source
- Used for filtering, storing a small amount of energy, and timing circuits



Integrated Circuits

- An **integrated circuit (IC, chip, microchip)** is a set of electronic circuits on one small flat piece of semiconductor material
 - Electronic components are **integrated** on the chip
 - Complex circuits can be **scaled down** and **mass-produced**
 - Used for a specific application



SECTION III

Microcontrollers

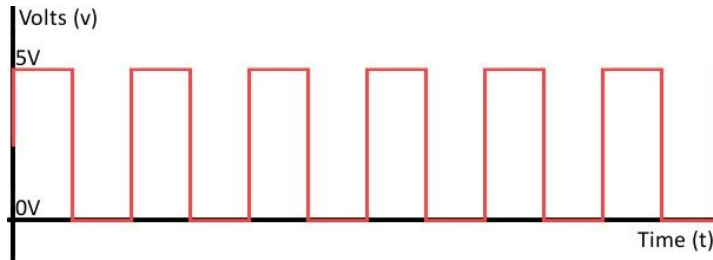
Microcontroller Reflection

- What does GPIO stand for?
- What are the main differences between digital and analog signals and when should you use one over the other?
- What is PWM and why is it important?



Digital Signals

- **Computers (and microcontrollers) transfer data** across wires/lines as **digital** (discrete) **voltage signals**
 - These signals are either a **HIGH** or **LOW** voltage
 - Contrasts from **analog signals** which can be values in a continuous range



This is a **digital signal** where the waveform is either **5V** or **0V** (two *discrete* values)

- **Digital signals are translated** to the **Binary** number system (1s and 0s)

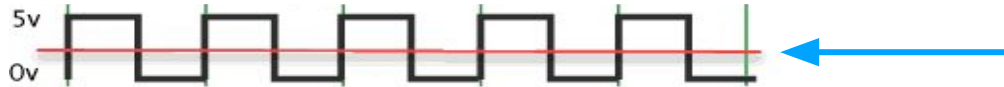
ADC and analogRead

- **analogRead** utilizes the analog-to-digital converter inside the microcontroller to **measure the real-world, analog signal** and **convert it to a digital signal**
 - The measurement resolution is 10 bits, which is why the function returns values from 0–1023



PWM

- The ESP32 board (the underlying AVR microcontroller) is a **digital source**, meaning it can only output a **HIGH** (5V) or **LOW** (0V) voltage
 - Then how does **analogWrite** output analog signals?
- **Pulse Width Modulation (PWM)** is an oscillating digital waveform that emulates an analog output
 - By oscillating a signal from **HIGH** to **LOW** quickly, the average voltage over time will be *between* **HIGH** and **LOW** - an analog value



The average value, the *analog value*, of this waveform is **2.5V**

SECTION IV

Communication Protocols

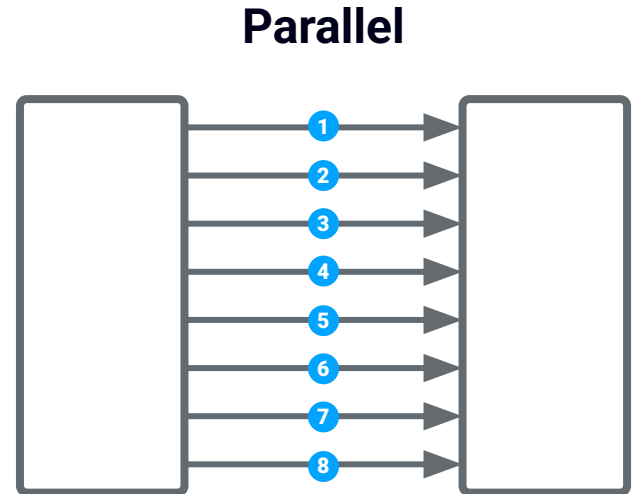
Communication Protocols Reflection

- What's the difference between synchronous and asynchronous protocols?
 - What synchronous protocols did we learn?
 - What asynchronous protocols did we learn?
- Why are clock signals important for SPI and I2C?



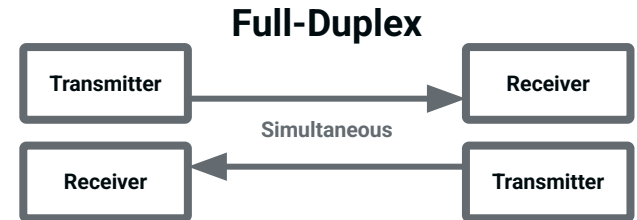
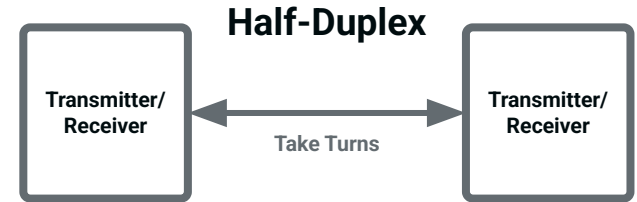
Serial vs Parallel Protocols

- **Serial** - bits are sent over a connection one by one to a device
 - Bits are sent in a specific order
 - The most common protocols are serial
- **Parallel** - multiple bits (often one byte) are sent simultaneously over a connection
 - This connection requires more wires, which takes up more space
 - Higher transmission rate



Transmission Modes

- **Simplex** protocols allow communication in only one direction
- **Half-Duplex** protocols allow communication in both directions but only in one direction at a time
- **Full-Duplex** protocols allow communication in both directions simultaneously

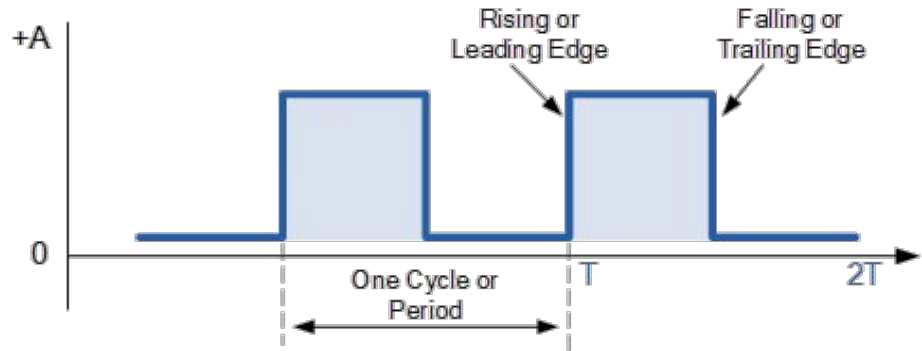


UART Protocol

- UART is a **communication protocol with certain characteristics**. It has rules for data transmission.
 - UART can be configured to **full-duplex, half-duplex, and simplex modes**
 - There are only **two lines** (electrical connections)
 - There is **no clock signal**/line hence the “asynchronous” part of the title
 - The data transmission speed is determined by a **baud rate**
 - It is quite **slow compared to other protocols**

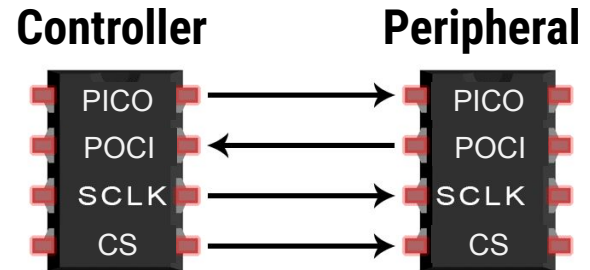
Clock Signals

- **Digital signals** are timed to a **clock signal** - most often, a square wave with a 50% duty cycle
 - For example, in a serial signal, one bit of data is sent per clock cycle
 - The bit might be timed to the rising or falling edge of the clock signal
- **Clock speed** determines the rate at which data is transmitted
 - measured as **frequency (Hz)**
 - inversely proportional to cycle (or period)

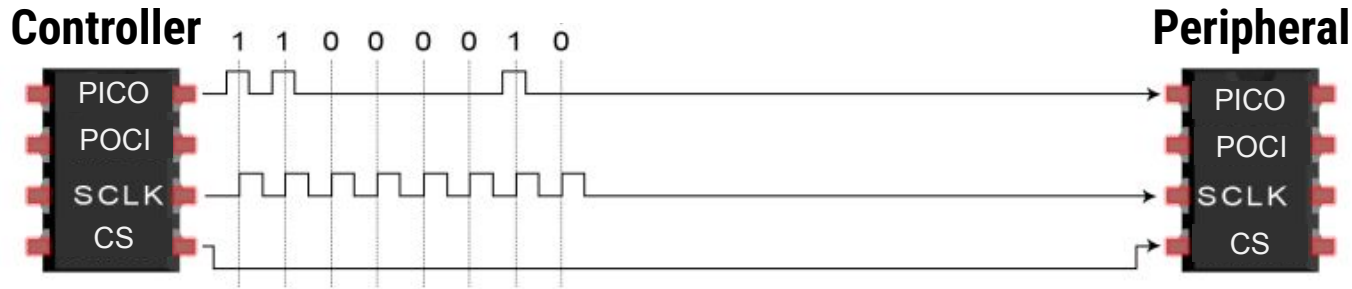


SPI Protocol

- **Serial Peripheral Interface** or **SPI** is a serial communication protocol which is **synchronized with a clock**
- Unlike UART, **data is transferred continuously** in SPI
 - There are no packets, start bits, stop bits, parity bits, or anything else
- One device acts as a **controller** to one or more **peripheral** devices
- Supports **full-duplex** communication
 - Bi-directional communication



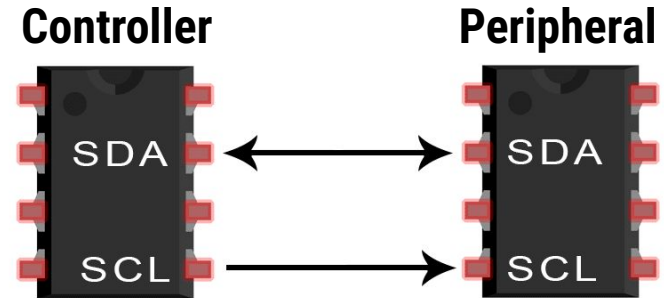
SPI Transmission



1. The controller first **selects a peripheral** by pulling a HIGH signal to LOW along the Chip Select (CS) line
2. The controller **sends data** to the selected peripheral along the Peripheral-In/Controller-Out (PICO) line
 - Bits in the PICO signal are **synchronized with the rising edge** of the oscillating clock signal
 - Alternatively, the peripheral sends data along the Peripheral-Out/Controller-In (POCI) line

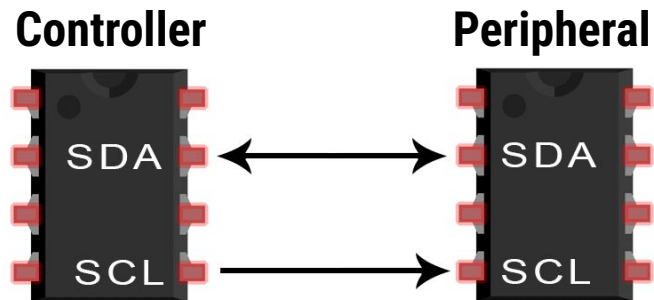
I²C Protocol

- **Inter Integrated Circuit** or I²C is another synchronized, serial protocol
- Unlike UART and SPI, I²C supports **multiple controllers**
- I²C features **acknowledgements** to confirm if messages are received and addressing for peripherals
- Supports **half-duplex** communication
 - One way communication



I²C Layout

- **SDA (Serial Data)** - the line for the controller and peripheral to send and receive data
- **SCL (Serial Clock)** - the line that carries the clock signal
- All peripherals can share the same two SDA and SCL lines



UART vs SPI vs I²C

Protocol	UART	SPI	I ² C
# of Lines	2	4+	2
# of Controllers	1	1	1+
# of Peripherals	1	1+	1+
Transmission Type	Full-Duplex	Full-Duplex	Half-Duplex
Error-Checking	✓	✗	✓
Speed	Slowest	Fastest	Slower

SECTION V

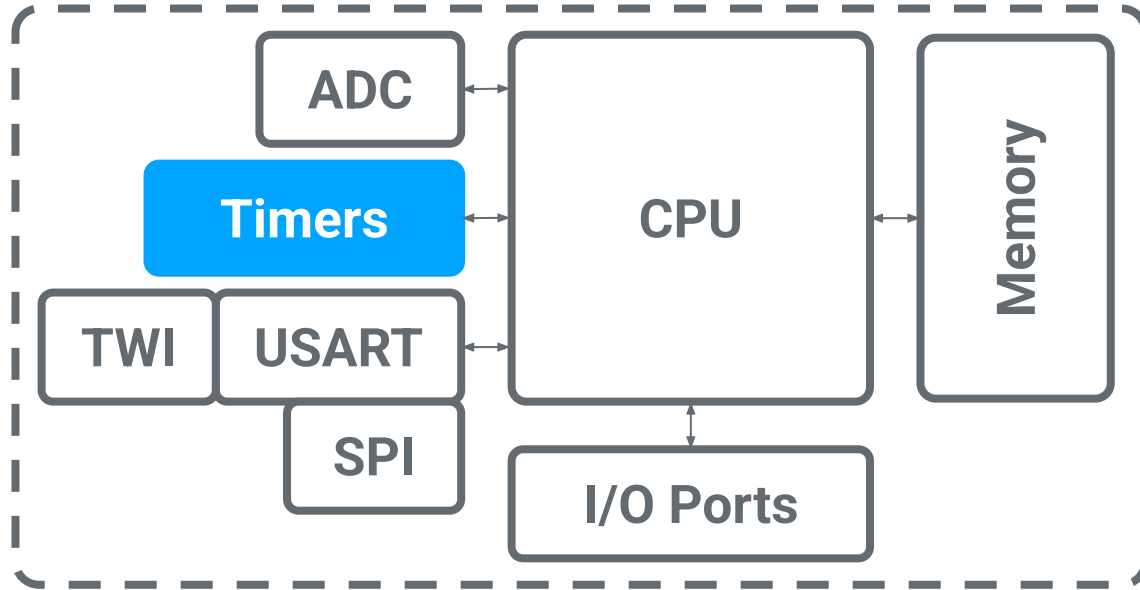
Timers & Interrupts

Timers & Interrupts Reflection

- What do timers do and what applications could they have?
- Why do we use interrupts and why can they be useful?
- What's the difference between hardware and software interrupts?



Timers



Timers are used by the microcontroller to **control the timing** of program execution or output signals and to **measure time**. Based on microcontroller clock cycle

Types of Timers

- Timer types can be distinguished by when the **timeout** triggers
 - A **timeout** is an event that occurs after a preset period of time
- A **one-shot timer** has a single timeout
 - For the timer to be run again, it must be **reloaded manually**
 - Ex) A countdown timer runs just once, stopping when it reaches zero
- A **periodic** (or **auto-reload**) **timer** has *periodic* timeouts that occur at a fixed interval
 - The timer **reloads automatically**
 - Ex) An alarm clock rings once every morning

Interrupts

- An **interrupt** is a request for the CPU to **halt the currently executing code** when an event occurs
 - The CPU suspends the current program to *handle* the event by executing a function called an **interrupt handler** or **interrupt service routine (ISR)**
 - When an interrupt halts the current code, a separate program / function can be called and run.
 - When the interrupt handler finishes execution, the CPU returns to the old program

Polling

- **Polling** is a method of periodically checking the status of a device
 - Ex) A person checking their phone for new messages every 10 minutes
 - We have been using polling in the `void loop()`
 - Polling loops cost additional CPU time
- Should we use interrupts instead of polling?
 - Interrupts signal *when* an event occurs whereas polling *checks* to see if an event occurred
 - Ex) A person reading their phone upon receiving a push notification
 - **Use interrupts when an event is urgent and/or infrequent**

Types of Interrupts

- A **software interrupt** is triggered by an instruction
 - Also called a **trap** or **exception**
 - Often caused by an illegal operation
 - Ex) Dividing by zero, Accessing memory without permission
- A **hardware interrupt** is triggered by devices external to the CPU
 - Also called an **external interrupt**
 - Caused when a device sends an **interrupt request (IRQ)** signal to the CPU
 - Ex) Timer timeout, button press, power failure

SECTION VI

PCB Design

PCB Design Reflection

- Why do we use PCBs instead of breadboards?
- What are common issues that can occur when creating/assembling a PCB?



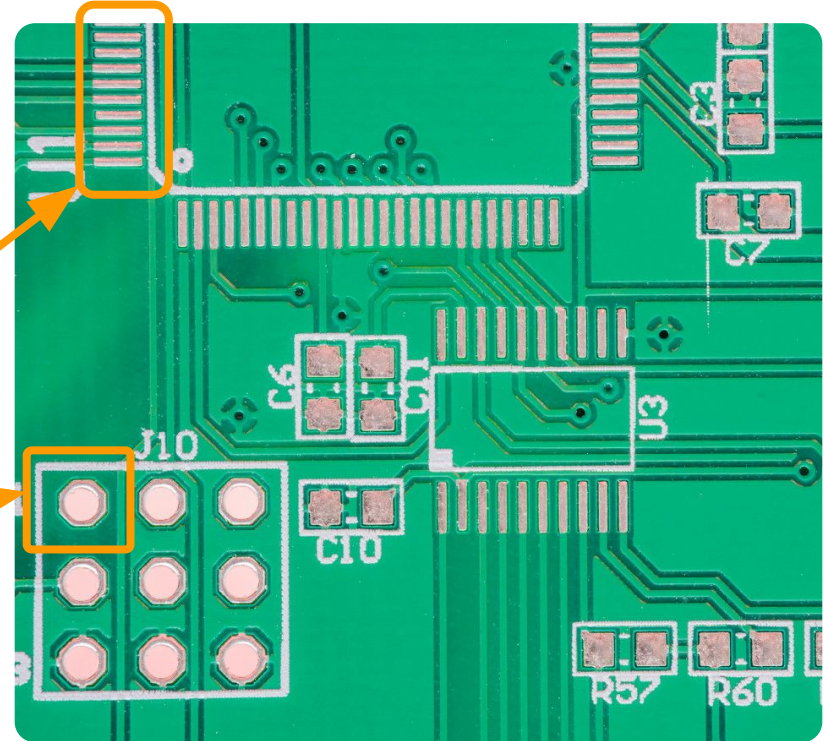
Printed Circuit Boards (PCBs)

- A **printed circuit board (PCB)** is an electronic assembly that uses copper conductors to **connect components**
 - It acts as a permanent map in placing and connecting electronic components
 - Made from **multiple, alternating layers of conductive** (usually copper) and **insulating material**



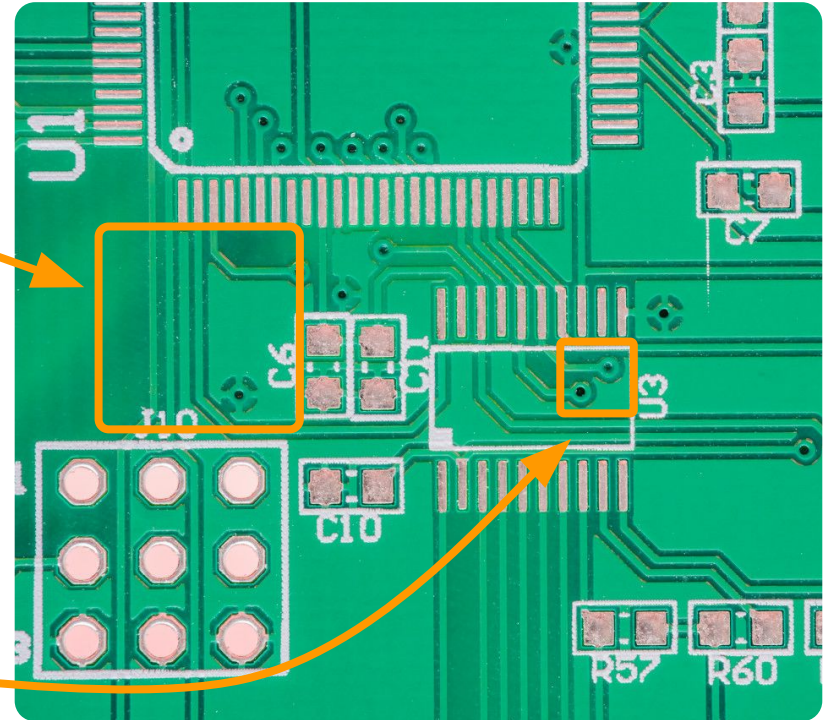
How are Components Connected?

- **Pads** are exposed copper surfaces which connect the leads of the electronic components to the board
 - Some pads are designed for **surface-mounted (SMD)** leads while others are for **through-hole (THT)** leads
 - Component **leads are soldered to the pads**



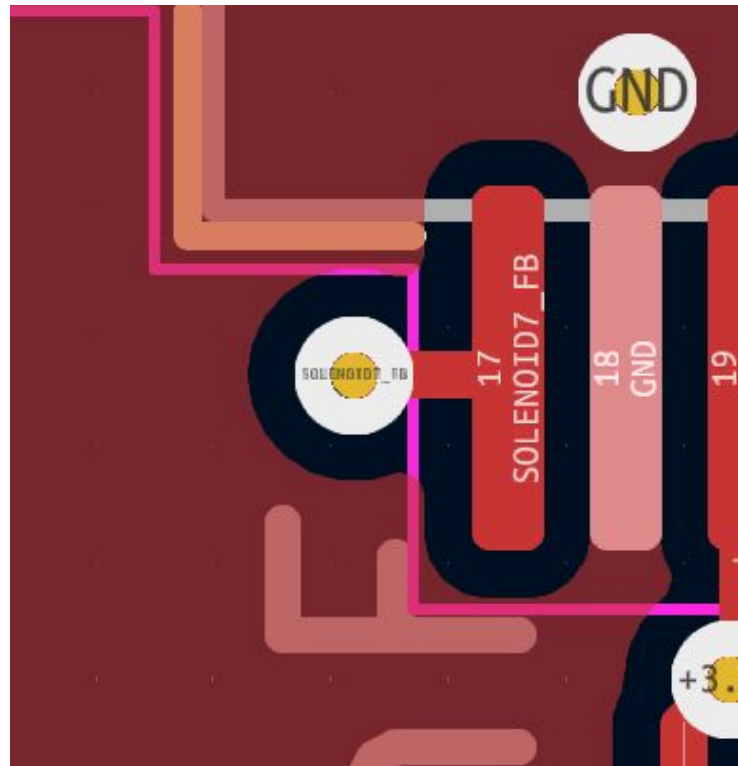
How are Components Connected? (Cont'd)

- **Traces** are **copper tracks** which connect pads
 - They are covered by a layer called the **solder mask** (colored green in the image)
- **Vias** are **conductive holes** drilled into the board to **connect different copper layers**



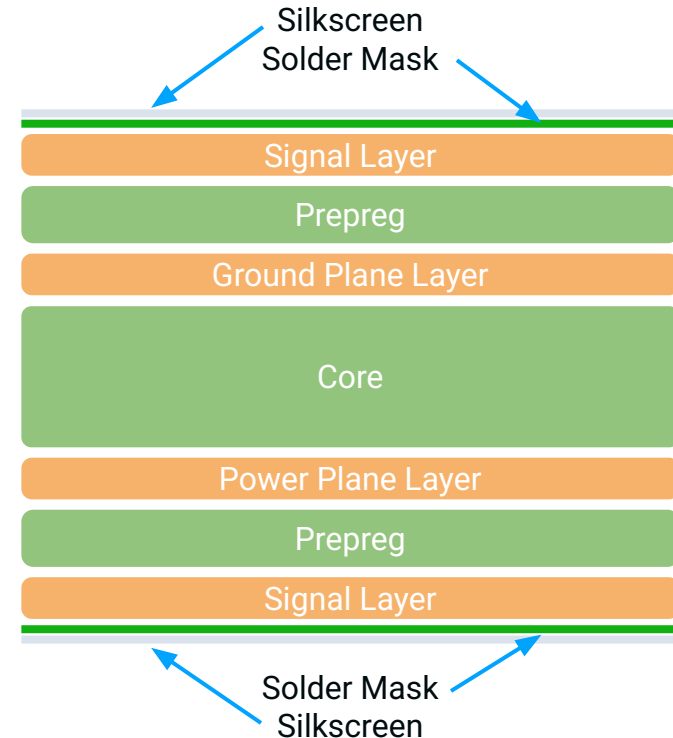
How are Components Connected? (Cont'd)

- A **plane** is an **inner conductive layer**
 - Used to create a ground point
- **Fills** are **large areas of copper** used for the same purpose as planes but can be integrated into the **same layer as traces**
 - The transparent shape surrounding the vias and the pads on the right is a Fill, that is used to ground the ground pad



PCB Stack-Up

- The **PCB Stack-Up** is the **arrangement** of PCB layers
- Most PCBs have multiple of the following layers:
 - **Copper layer**
 - Signal or routing layer
 - Ground/Power plane layer
 - **Insulation layer**
 - Core
 - Prepreg
 - **Solder mask layer**
 - **Silkscreen layer**



SECTION VII

Closing Remarks

Motivation for Studying

- The **capstone project** makes up a significant part of the **hardware design process**
 - Breadboard → PCB
- These fundamental concepts are likely to appear during **technical interviews**
- EECS and **embedded systems** are concepts that are varied and have lots of depth, much more than what we could cover in OPS
 - We encourage you to do your own research and get involved with anything you find interesting!
 - You could even explore topics related to what we covered this year in OPS!

Upcoming Events

Week	Suggested Activity	Workshop
1	Capstone PCB Design Attend Lecture: Capstone Project: Getting Started	
2		Capstone Rover Assembly
3	Attend Lecture: Senior Design Project Showcase	
4		Capstone PCB Assembly
5	Start Working on Capstone Programming Independently (Lecture Tentative)	
6		Capstone Programming
7	Attend Lecture: End of the Year Review/Reflection <i>OPS Picnic Social!</i>	
8	Suggested Capstone Project Submission Date: May 22nd	
9	Rover Competition	
10	OFFICIAL Capstone Project Submission Deadline: June 5th 2026 at 11:59PM	

Thank You!

Add image here

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